

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=butter

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.08225840900715364	0.01	2.681802621264837e-12	0.1	1.3006957963952123e-20	0.1	I(1)	1.0	butter	silpo	promo_controlled	nan	nan
prozorro	0.056404226942647726	0.01	3.518946761497874e-22	0.1	1.9269184182026386e-13	0.1	I(1)	1.0	butter	silpo	promo_controlled	nan	nan
retail	2.2989340931225115e-06	0.1	2.5595238291844733e-09	0.1	8.70860721185971e-05	0.04166666666666662	I(0)	0.0	butter	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo	promo_controlled	0.46279135586174147	220.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo	promo_controlled	8.244750384971287e-07	41.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	butter	silpo	promo_controlled	0.9173535311228344	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
butter	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.21277642294253607	0.27405320585301884	-0.8083252331134957	1.2523745817378955e-26	0.787381486847121	0.20112635504345666	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...
butter	silpo	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	41	0.2001943602714764	-0.4257756633486405	nan	nan	nan	nan	nan	unreliable	1	bounds_lower_p=0.1096; bo...	Primary chain is strictly...
butter	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	39	0.04860976737983126	-0.12648415712694783	-0.8756967529857956	2.214481682557665e-08	0.5941357916590037	0.5501601441288803	nan	unreliable	1	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
NARDL	producer_to_prozorro	prozorro	producer	0.3043740439991276	0.5287391650637272	4.2203813853019655e-84	0
ARDL	prozorro_to_retail	retail	prozorro	0.01144894530681835	0.09344174279103216	0.6443904339549587	1
NARDL	prozorro_to_retail	retail	prozorro	0.5591712653384464	0.046665474894684277	0.8050949849331446	1

Table: SeriesUsed

	date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
	2021-02-01	31.78268739915295	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-02	31.74153855497394	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-03	31.71205395008976	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-04	31.69228954461795	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-05	31.68030129867603	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-06	31.67414517238153	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-07	31.67187712585198	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-08	31.6715531192049	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-09	31.71501589339797	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-10	31.82710410052748	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-11	31.98036756741884	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-12	32.14735612089748	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-13	32.30061958778885	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-14	32.41270779491835	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-15	32.45617056911143	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-16	32.45341900628176	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-17	32.44522030312711	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-18	32.43165843764893	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-19	32.41281738784873	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-20	32.38878113172797	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-21	32.35963364728814	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-22	32.32545891253073	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-23	31.91148084818262	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-24	30.90284056637734	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-25	29.53949783222626	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-26	28.06141241084077	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-27	26.70854406733228	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-02-28	25.72085256681216	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-01	25.33829767439181	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-02	25.36591783597624	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-03	25.44026511447978	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-04	25.54856970052784	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-05	25.67806178474581	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-06	25.81597155775909	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-07	25.94952921019307	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-08	26.06596493267315	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-09	26.18214222460376	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-10	26.31503339825227	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-11	26.45203186353847	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...
	2021-03-12	26.58053103038216	nan	nan	butter	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

pair	lag	corr
producer_to_prozorro	1	0.07873337531291687
producer_to_prozorro	2	0.08283043430904553
producer_to_prozorro	3	0.08251800718380085
producer_to_prozorro	4	0.08422490303011267
producer_to_prozorro	5	0.0907848031216401
producer_to_prozorro	6	0.08185920607839353
producer_to_prozorro	7	0.08099913022176998
producer_to_prozorro	8	0.08418185337829556
producer_to_prozorro	9	0.08011971167494186
producer_to_prozorro	10	0.08071177659888301
producer_to_prozorro	11	0.0855520813606708
producer_to_prozorro	12	0.08891786146062777
producer_to_prozorro	13	0.0906935444584109
producer_to_prozorro	14	0.09336004953876562
producer_to_prozorro	15	0.09806722582073225
producer_to_prozorro	16	0.10411102969445078
producer_to_prozorro	17	0.11083781027515911
producer_to_prozorro	18	0.11554030327110705
producer_to_prozorro	19	0.11683243707057434
producer_to_prozorro	20	0.11514579002515733
producer_to_prozorro	21	0.11293989092313386
producer_to_prozorro	22	0.11348584116046659
producer_to_prozorro	23	0.11614446866624657
producer_to_prozorro	24	0.1193793796780393
producer_to_prozorro	25	0.12118388938915296
producer_to_prozorro	26	0.11984324455041497
producer_to_prozorro	27	0.11226973352035723
producer_to_prozorro	28	0.1007561552537856
producer_to_prozorro	29	0.09278063269691687
producer_to_prozorro	30	0.09120495228803738
prozorro_to_retail	1	0.027215636352813634
prozorro_to_retail	2	-0.1306328650069521
prozorro_to_retail	3	0.1914449373424407
prozorro_to_retail	4	0.1048257656843792
prozorro_to_retail	5	-0.0807571026088193
prozorro_to_retail	6	-0.02993875084689255
prozorro_to_retail	7	-0.00117040461907769
prozorro_to_retail	8	-0.26096787569451063
prozorro_to_retail	9	-0.16753726563798146
prozorro_to_retail	10	0.02498505897512193

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
butter	silpo	promo_controlled	daily	I(1)	I(1)	I(0)	1793	230	48	220	41	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

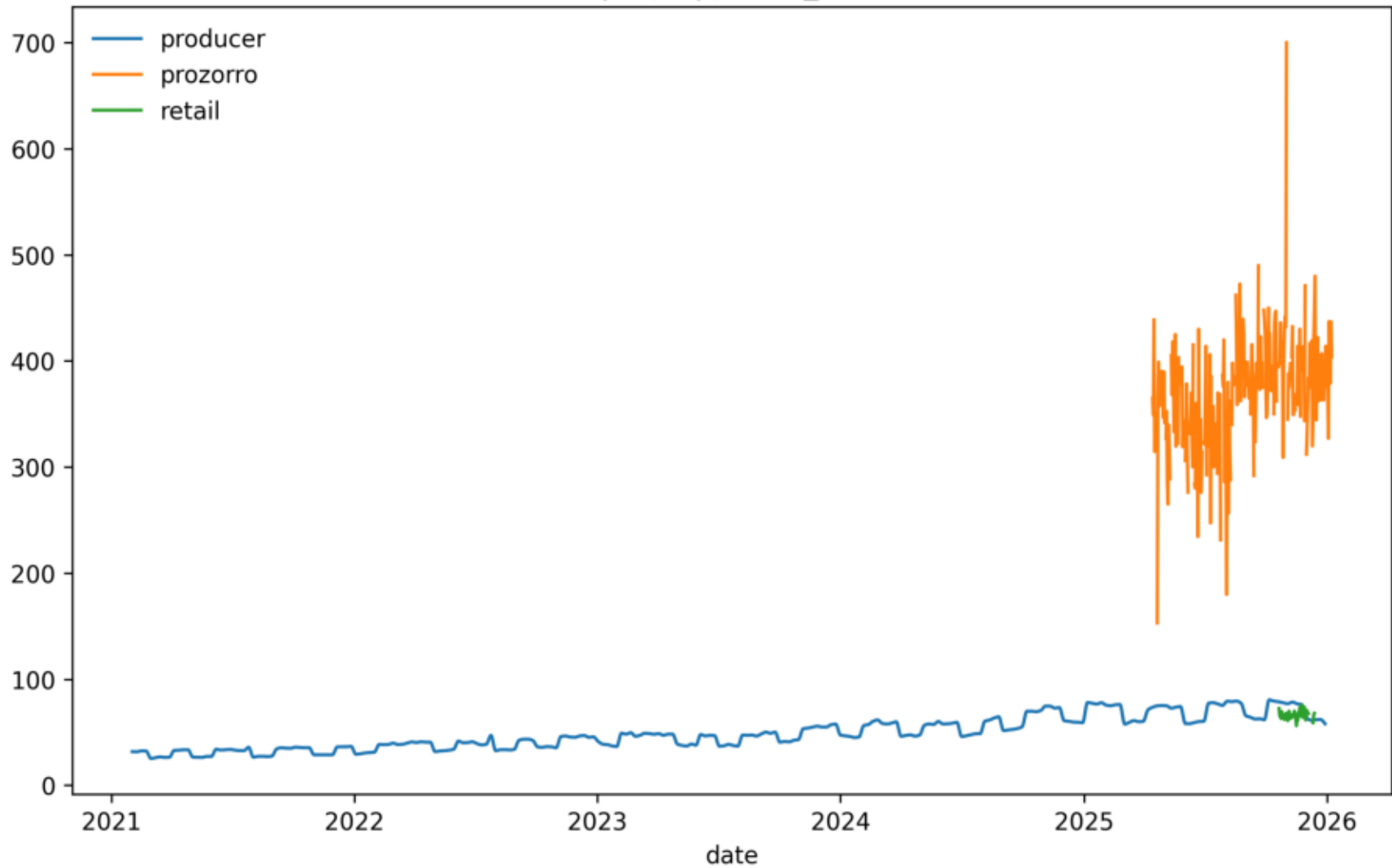
horizon	mult_pos		mult_neg	link
0	-0.5437076425441756	-0.3309312196016395	producer_to_prozorro	
1	-0.4804346363943778	-0.29241968977462707	producer_to_prozorro	
2	-0.4877979195156134	-0.2969014002986514	producer_to_prozorro	
3	-0.48694103058899735	-0.2963798492381931	producer_to_prozorro	
4	-0.4870407495079181	-0.2964405438116616	producer_to_prozorro	
5	-0.48702914489676974	-0.29643348058904007	producer_to_prozorro	
6	-0.4870304953626729	-0.2964343025589618	producer_to_prozorro	
7	-0.4870303382046146	-0.2964342069036797	producer_to_prozorro	
8	-0.48703035649360304	-0.2964342180353924	producer_to_prozorro	
9	-0.48703035436525466	-0.2964342167399592	producer_to_prozorro	
10	-0.4870303546129374	-0.2964342168907129	producer_to_prozorro	
11	-0.48703035458411376	-0.29643421687316923	producer_to_prozorro	
12	-0.4870303545874681	-0.2964342168752108	producer_to_prozorro	
13	-0.4870303545870777	-0.29643421687497323	producer_to_prozorro	
14	-0.4870303545871231	-0.2964342168750009	producer_to_prozorro	
15	-0.48703035458711785	-0.2964342168749977	producer_to_prozorro	
16	-0.48703035458711846	-0.29643421687499805	producer_to_prozorro	
17	-0.4870303545871184	-0.296434216874998	producer_to_prozorro	
18	-0.4870303545871184	-0.29643421687499805	producer_to_prozorro	
19	-0.4870303545871184	-0.296434216874998	producer_to_prozorro	
20	-0.4870303545871184	-0.29643421687499805	producer_to_prozorro	
0	0.060095225918782094	0.011485458538950832	prozorro_to_retail	
1	0.06495635205286496	0.01241452173510348	prozorro_to_retail	
2	0.06534957043136644	0.012489674017401843	prozorro_to_retail	
3	0.0653813780185235	0.012495753114666745	prozorro_to_retail	
4	0.06538395094658604	0.01249624485520244	prozorro_to_retail	
5	0.065384159071735	0.012496284632285106	prozorro_to_retail	
6	0.06538417590705903	0.012496287849868752	prozorro_to_retail	
7	0.06538417726887492	0.01249628811014034	prozorro_to_retail	
8	0.06538417737903274	0.01249628813119381	prozorro_to_retail	
9	0.06538417738794344	0.012496288132896834	prozorro_to_retail	
10	0.06538417738866423	0.012496288133034593	prozorro_to_retail	
11	0.06538417738872254	0.012496288133045735	prozorro_to_retail	
12	0.06538417738872726	0.012496288133046637	prozorro_to_retail	
13	0.06538417738872764	0.01249628813304671	prozorro_to_retail	
14	0.06538417738872766	0.012496288133046715	prozorro_to_retail	
15	0.06538417738872768	0.012496288133046717	prozorro_to_retail	
16	0.06538417738872768	0.012496288133046717	prozorro_to_retail	
17	0.06538417738872768	0.012496288133046717	prozorro_to_retail	
18	0.06538417738872768	0.012496288133046717	prozorro_to_retail	

Table: VECM_IRF

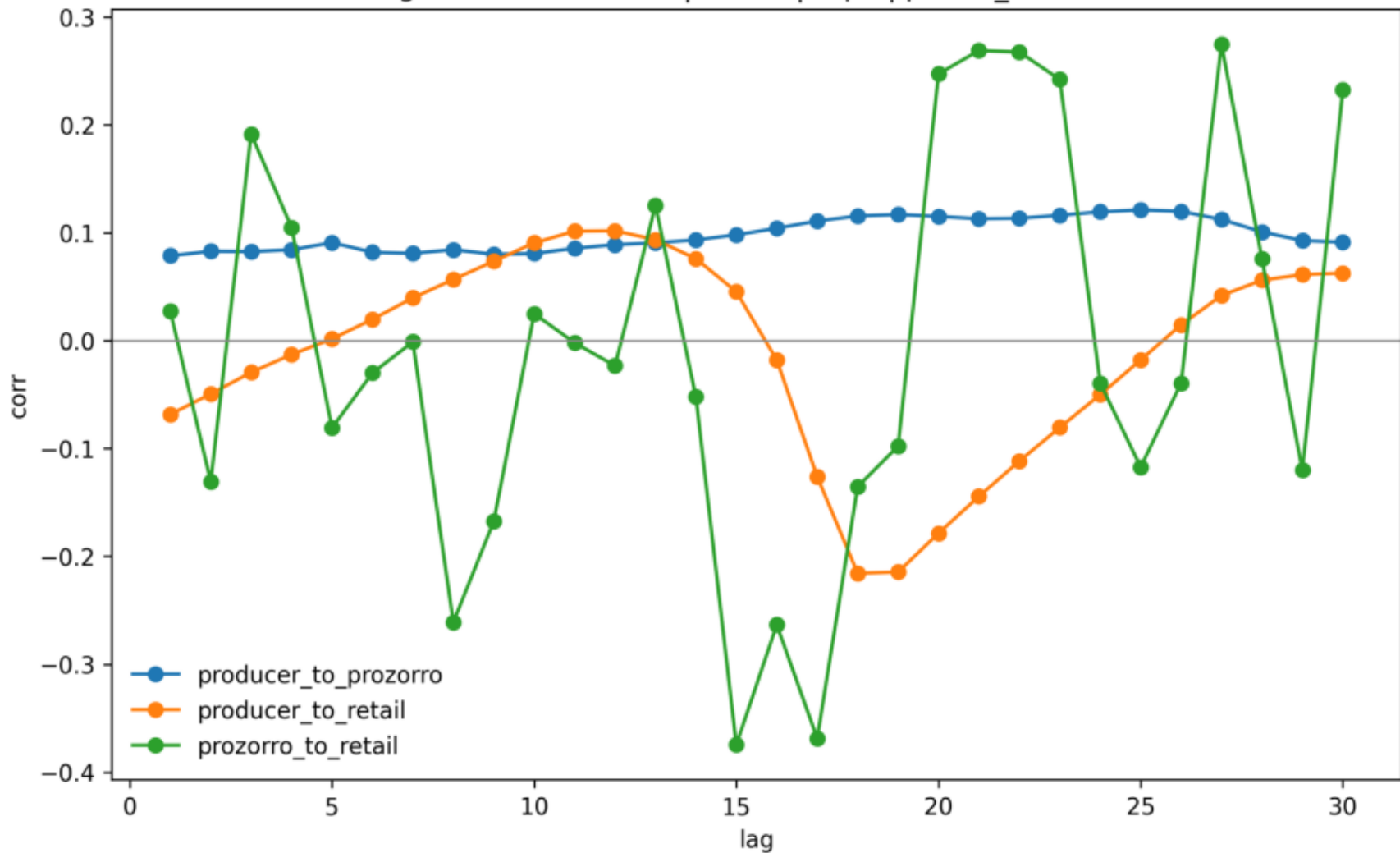
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time_series_promo_controlled.png

butter | silpo | promo_controlled



Lag-Correlation Profile | butter | silpo | promo_controlled



NARDL Dynamic Multipliers | butter | silpo | promo_controlled

